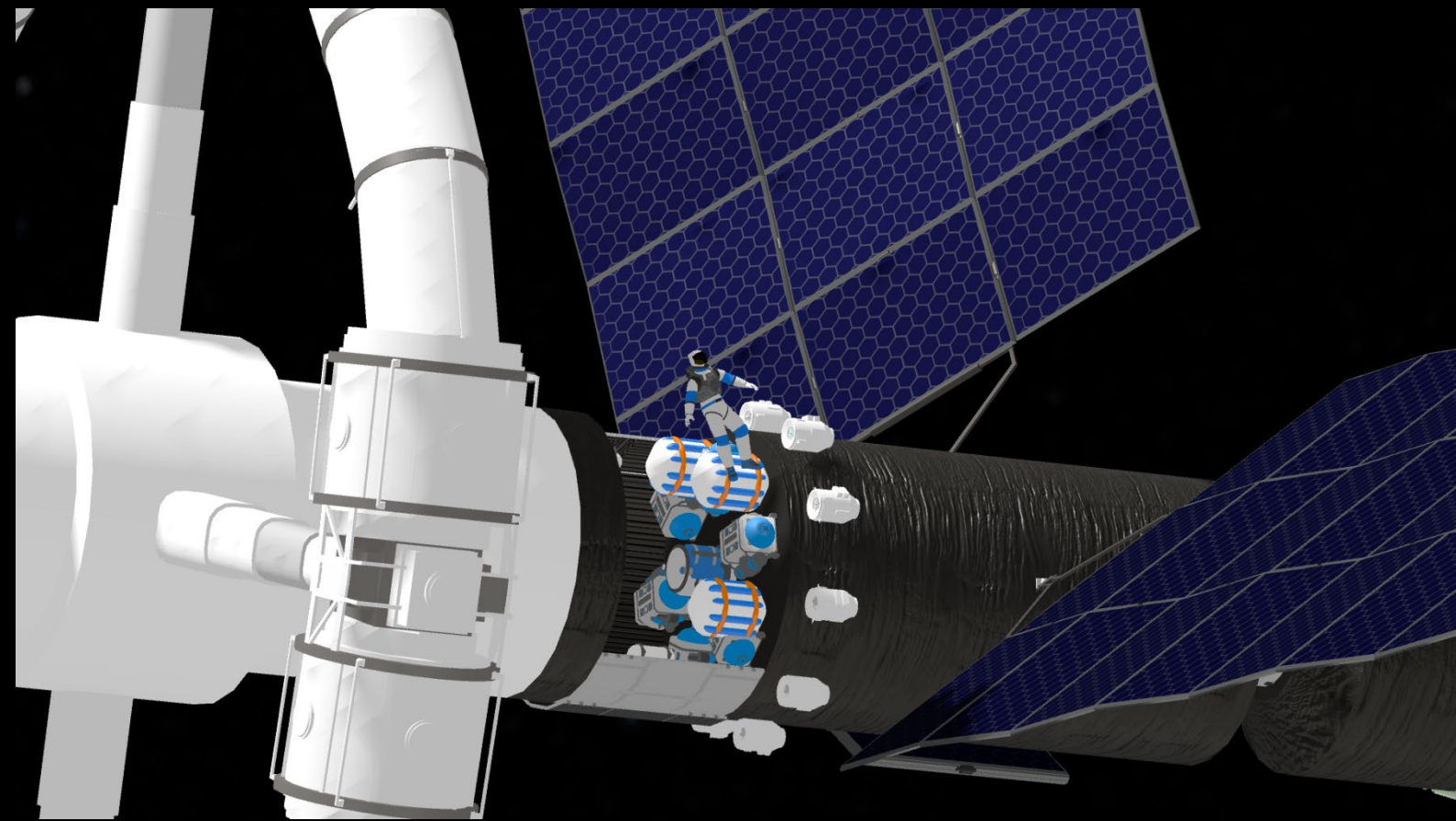




Droodism User Manual

Manual Version:3.0

Mod Version: 0.795

By SatelliteTorifune

0. What is Doodism?

Doodism is a Mod for Juno: New Origins which features life support system for astronauts and tourists (aka Dood) in game.

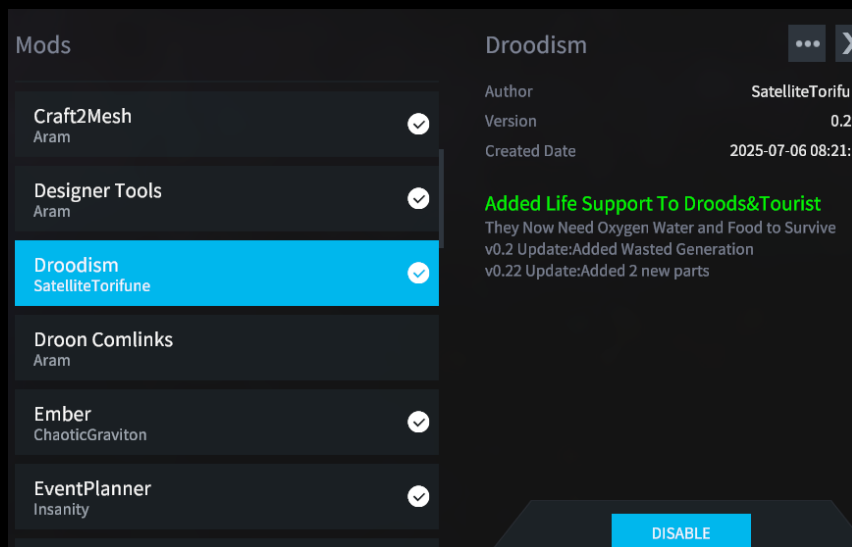
When designing your own manned spacecraft, the life support system becomes a necessary part to consider. Now your crew needs food, water, and oxygen to survive during the voyage to space. You will also need to think about how to deal with the waste.

Doodism provides multiple paths for handling life support resources. From simply attaching a food pack, to using a photobioreactor, or just putting your crew into hibernation, there will always be a solution.

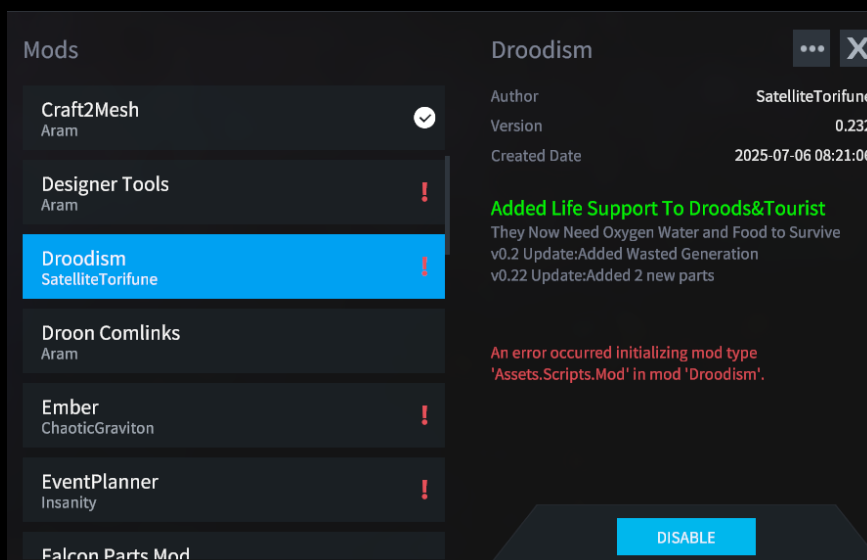
1. How to Install This Mod

Droodism requires the Juno Harmony Mod as a dependency which you can install [here](#).

After Downloading the Juno Harmony Mod, open mod manager and enable Juno Harmony Mod first, then enable Droodism.



successfully



failed



2. Droids

This mod added a new modifier for droid called. “SupportLife”, it is a core part of this mod. In the Designer, you can allocate the resources your crew will carry through. Sliders, Select Droid’s role, their life-long radiation dose.

The more resources they will carry, the more mass this droid will be; however, it does not affect the capacity of the waste.

You can also decide what kind of parachute to

bring for this drood, the parachute's detailed config data, like the auto-deployment height and the fully deployment height.



A new inspector panel has been added to droods for players to check data, like resources' amount, current radiation dose rate, and total mission time.

Droods have separate roles,



Pilots could reduce the damage caused by Over G by 40% and plant flags on surface of a plant.

Engineers could fix damaged parts during flight.

Scientists have no special abilities yet...

In the atmosphere of Planets with breathable environment like Droo Earth Kerbin Laythe Nebra, Drood will not consume internal oxygen. , and it will be filled up automatically.

For Configuring your own breathable planets or doing overrides, you could find the config list here,



AppData > LocalLow > Jundroo > SimpleRockets 2 > UserData > DroodismConfig > Planets				
授予访问权限 新建文件夹				
名称	修改日期	类型	大小	
 BreathablePlanets.xml	2026/05/27 15:40	XML 源文件	1 KB	

```
<BreathablePlanets>  
  <BreathablePlanet>Droo</BreathablePlanet>  
  <BreathablePlanet>Earth</BreathablePlanet>  
  <BreathablePlanet>Nebra</BreathablePlanet>  
  <BreathablePlanet>Kerbin</BreathablePlanet>  
  <BreathablePlanet>Laythe</BreathablePlanet>  
</BreathablePlanets>
```



Tourists will consume more resources than normal astronauts.



When running, Droods will consume more resources than standing still or walking.

You can get Droods abroad a craft for refilling the resources and removing the waste.



When running out of resources or when the waste is full, the Droid will take damage based on the type of resource or waste. For instance, when running out of oxygen or when Carbon Dioxide is full, it takes only minutes to kill a Droid. When running out of food, it takes days before your crew passes away.

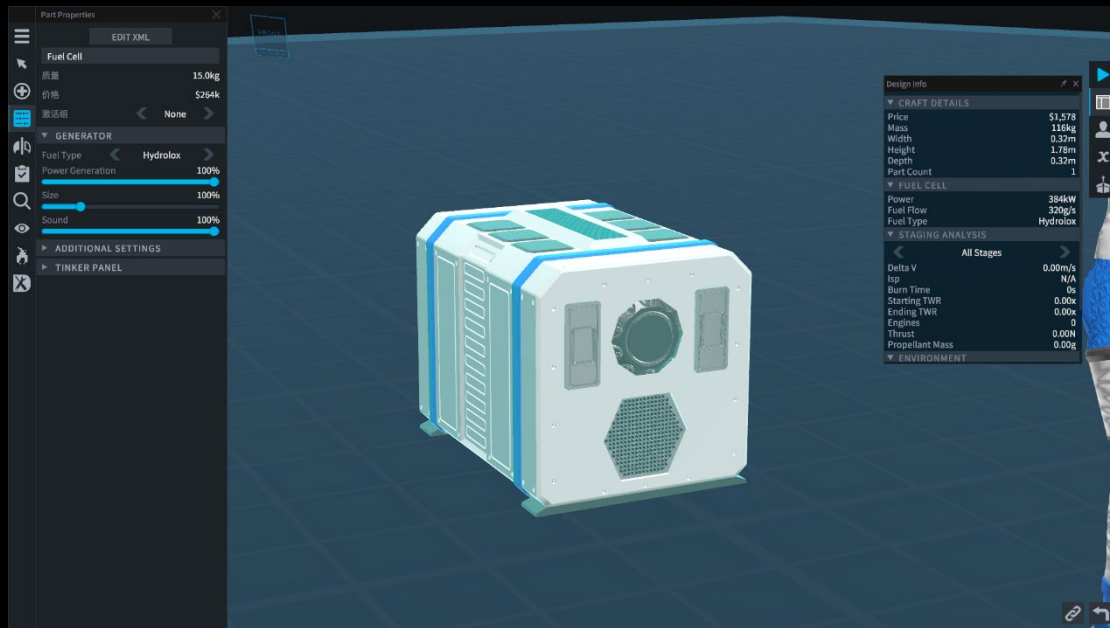
Also, running will kill a crew member more quickly if they are out of resources.

3. Parts Introduction

Droodism added many new parts for players to create their own craft with life support system.



You can find these new parts in the new category called "Life Support."



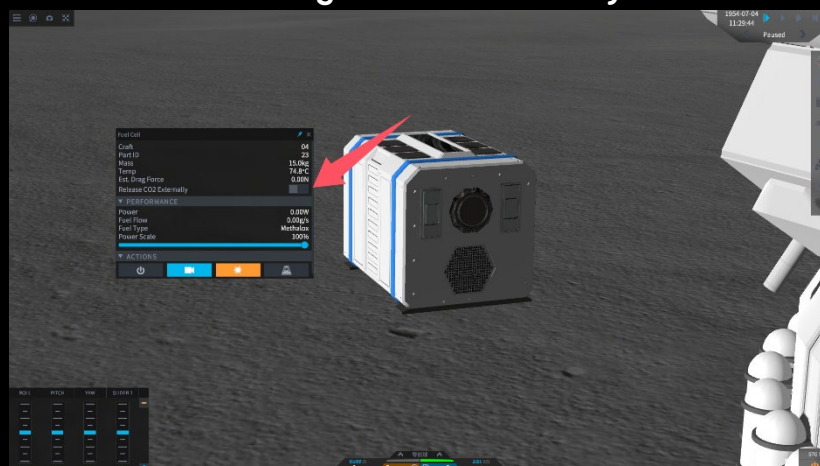
The stock part: Fuel cells can now generate resources while working depending on their fuel type.

Hydrolox now generates water and oxygen.

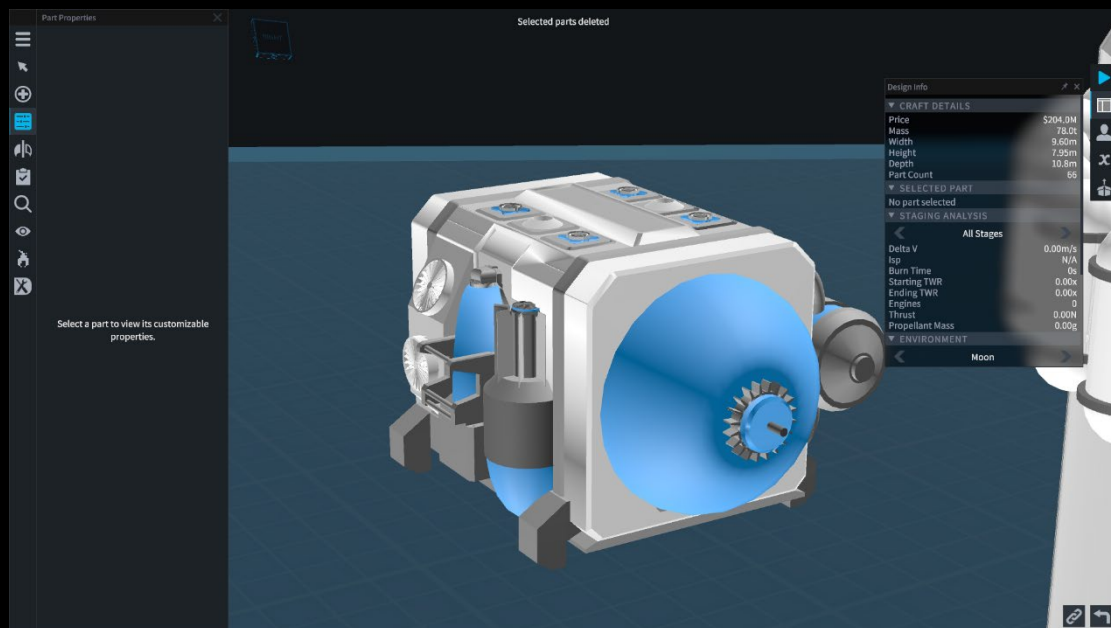
Methalox now generates water, Carbon Dioxide

Kerolox and Jet now generate Carbon Dioxide

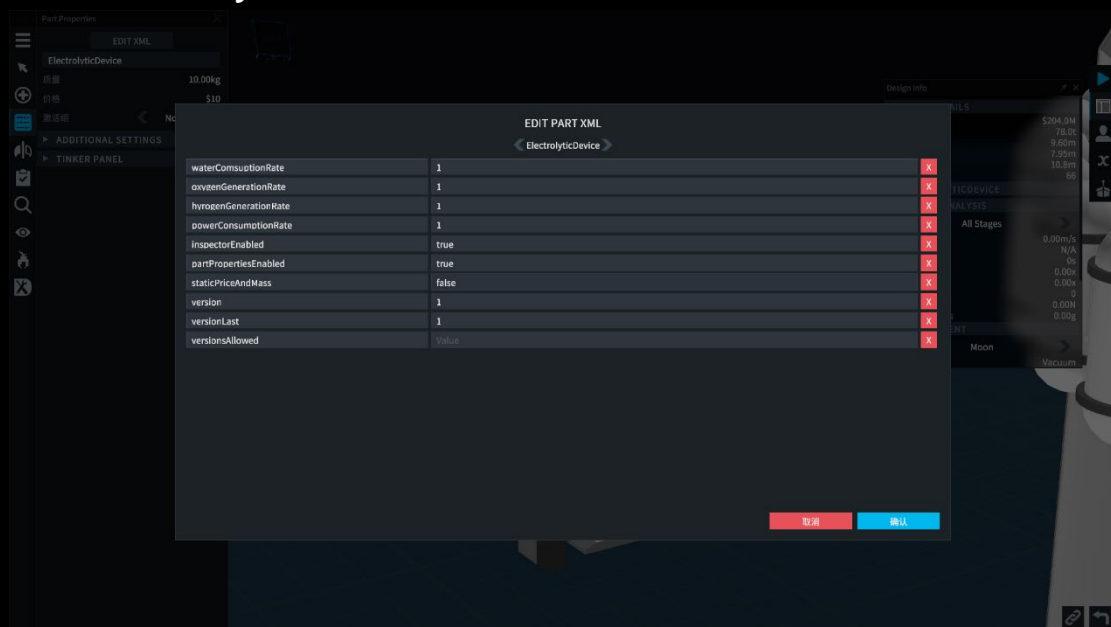
For these Carbon Dioxide-generating fuel cells, you can toggle to choose to keep the Carbon Dioxide or just release the gas externally.



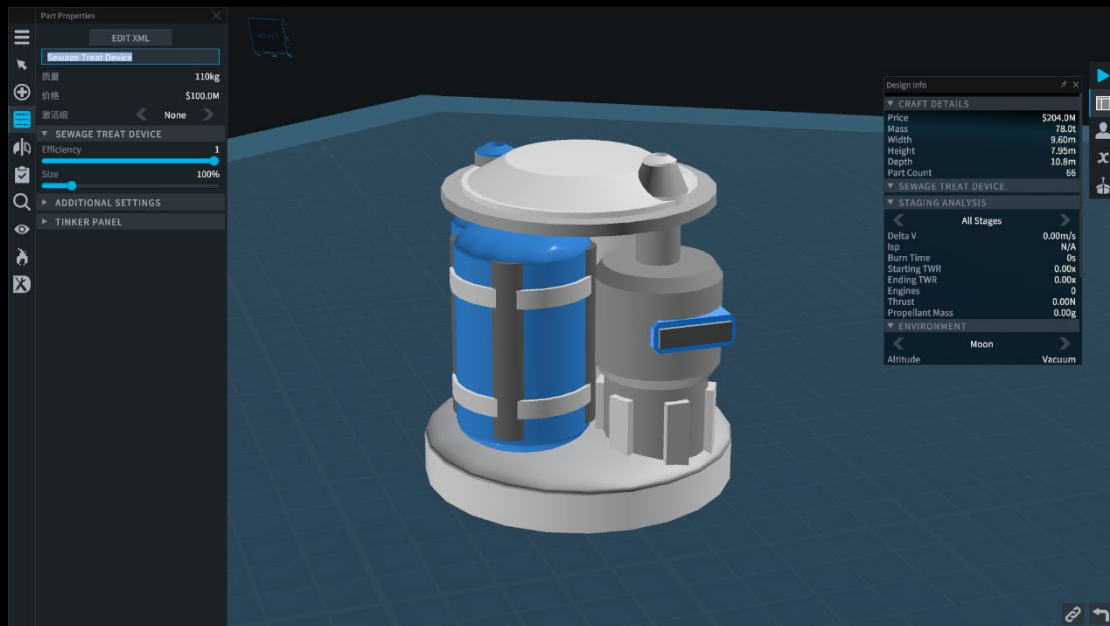
Electrolytic Device



This part does a very simple thing: turning water into oxygen and liquid hydrogen consuming electricity



You can edit the properties of this part. By editing the hidden properties.

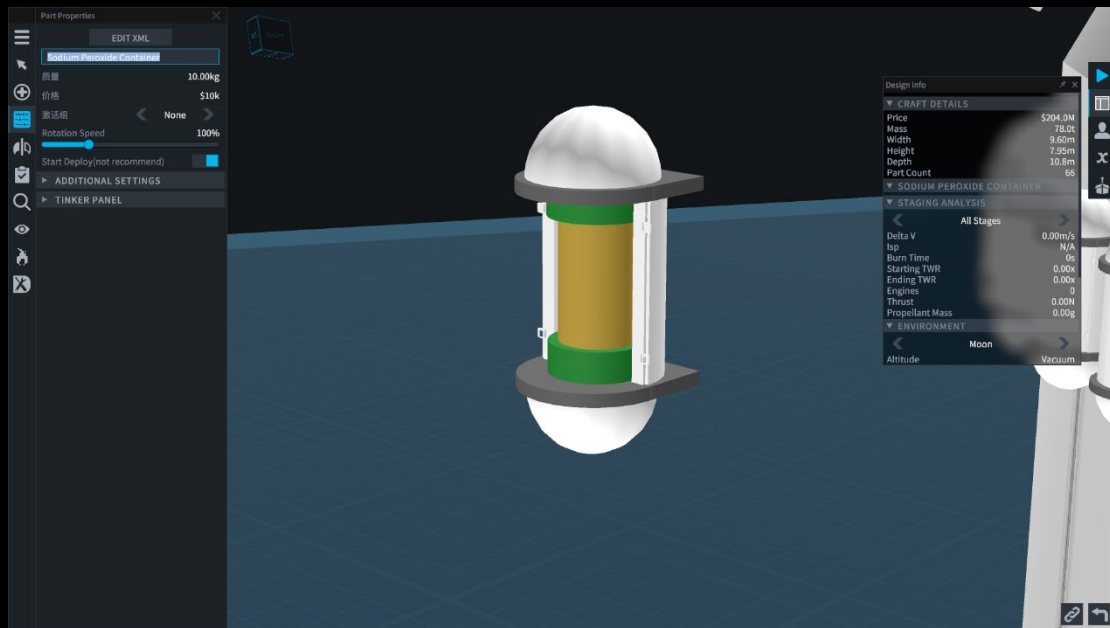


Sewage Treat Device

Your crew and some other parts will generate wasted water, wanna get rid of it? This is your choice.

Sewage Treat Device could turn wasted water into clean water ,consuming electricity, and you can edit the efficiency and size of the part.

Your crew and some other parts will generate wasted water, wanna get rid of it? This is your choice



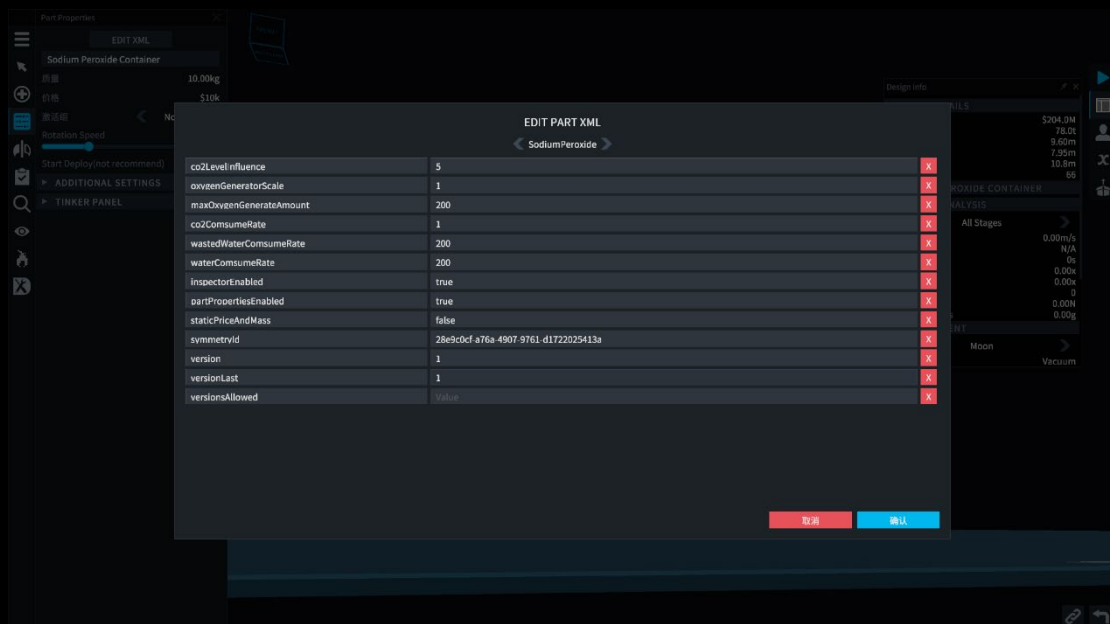
Sodium Peroxide Container

This part is what you need when there is too much Carbon Dioxide or wasted water, unlike other parts, this part does not consume electricity, it absorbs Carbon Dioxide and wasted water and gives out oxygen. The speed of releasing oxygen is based the level of current Carbon Dioxide.

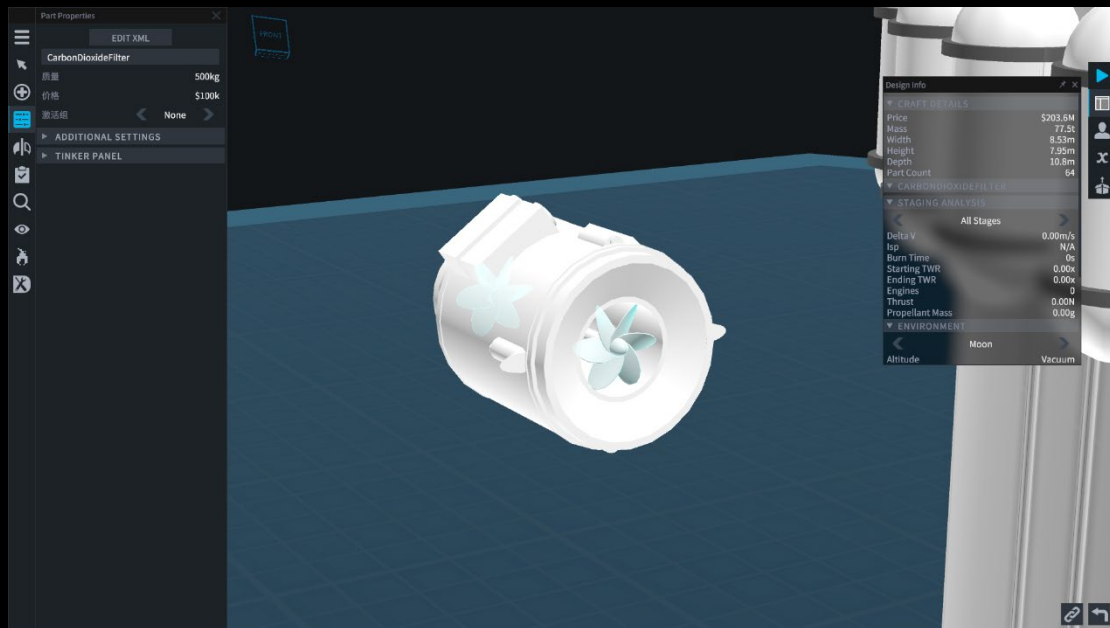
It is suggested that bring one or two of this part and attach it to your craft in case of emergency, however, this part is not reusable, and will become deadweight when it's used up



You can tell a Sodium Peroxide Container's states by checking from its inspector panel, the progress bar will tell you how far the generation reaction go.



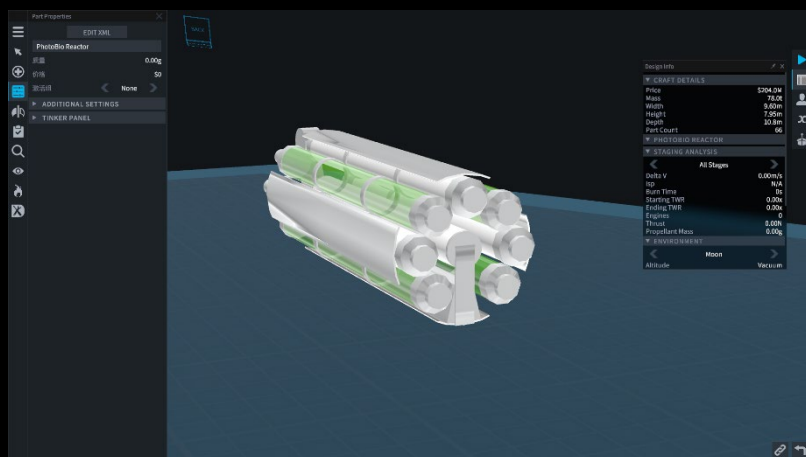
you can edit more properties here.



CarbonDioxide Filter

This part simply throws Carbon Dioxide in your craft to external environment, consuming electricity.

Just bring one, in case your craft's Carbon Dioxide level is too high for your crew to survive

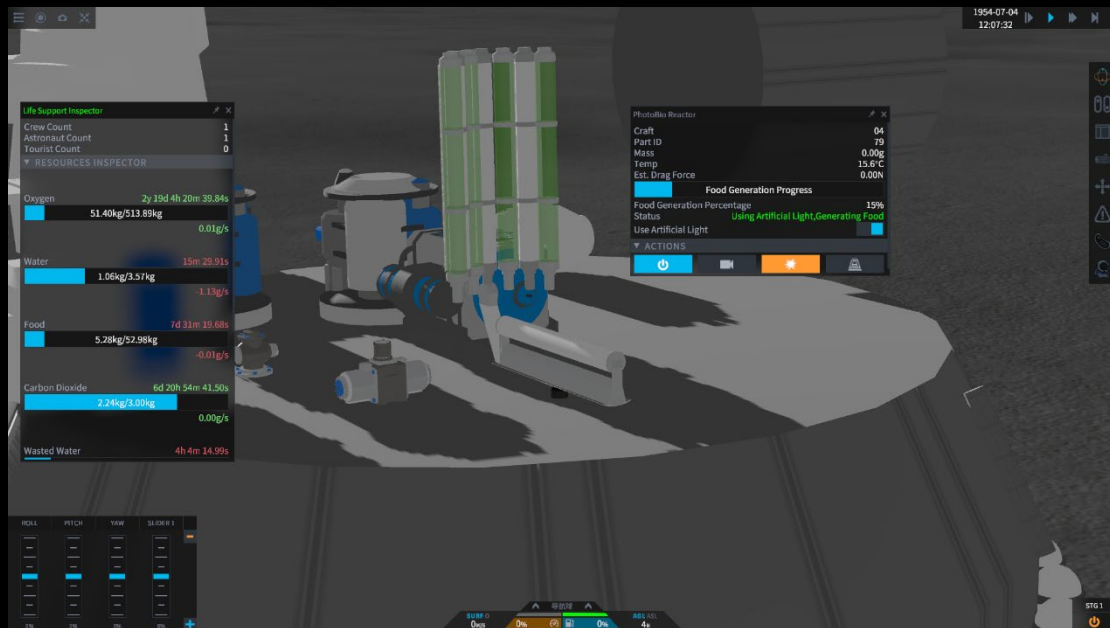


PhotoBio Reactor

This is rather a complex part, this part absorbs Carbon Dioxide and water, and gives out oxygen and wasted water, and solid waste will booster this process.



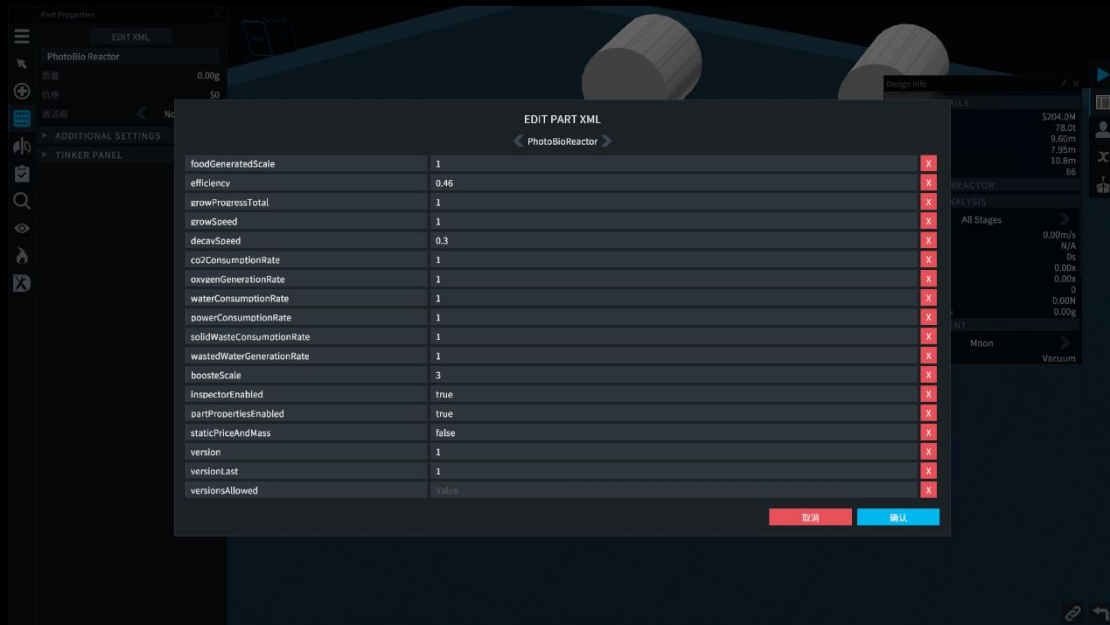
When provided with enough Carbon Dioxide and water, and with enough solar radiation, the part will gain progress in generating food. The efficiency is related to the solar radiation's pointing efficiency. However, it's also possible to use



electricity to generate artificial light, making it possible for this part to work even in the dark.

When using artificial light, the efficiency of this part is 100%

Please note that this part can not self-rotate toward the Sun like solar panel array.

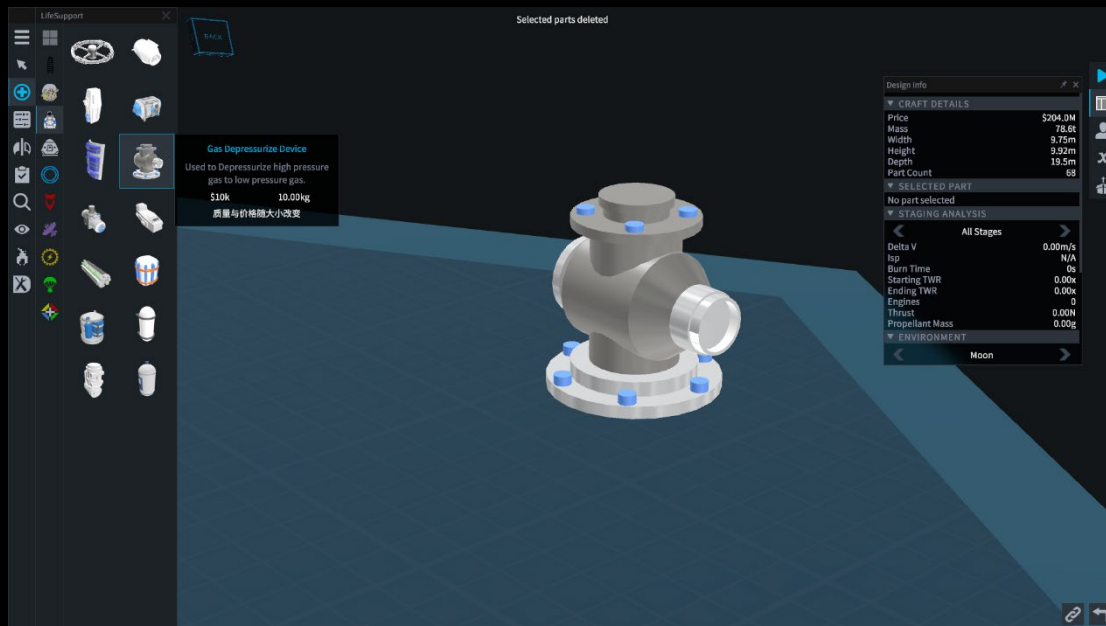


you can edit more properties here.

You may notice that oxygen gas and Carbon Dioxide have very low density. How did people in real space programs deal with this problem? The answer is high-pressure gas.

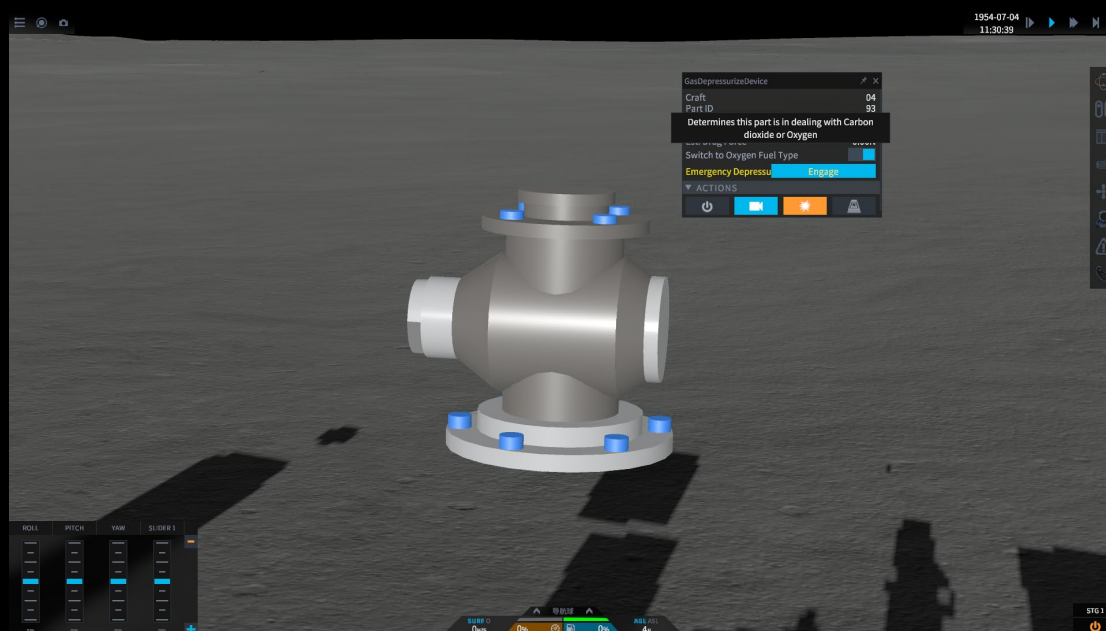


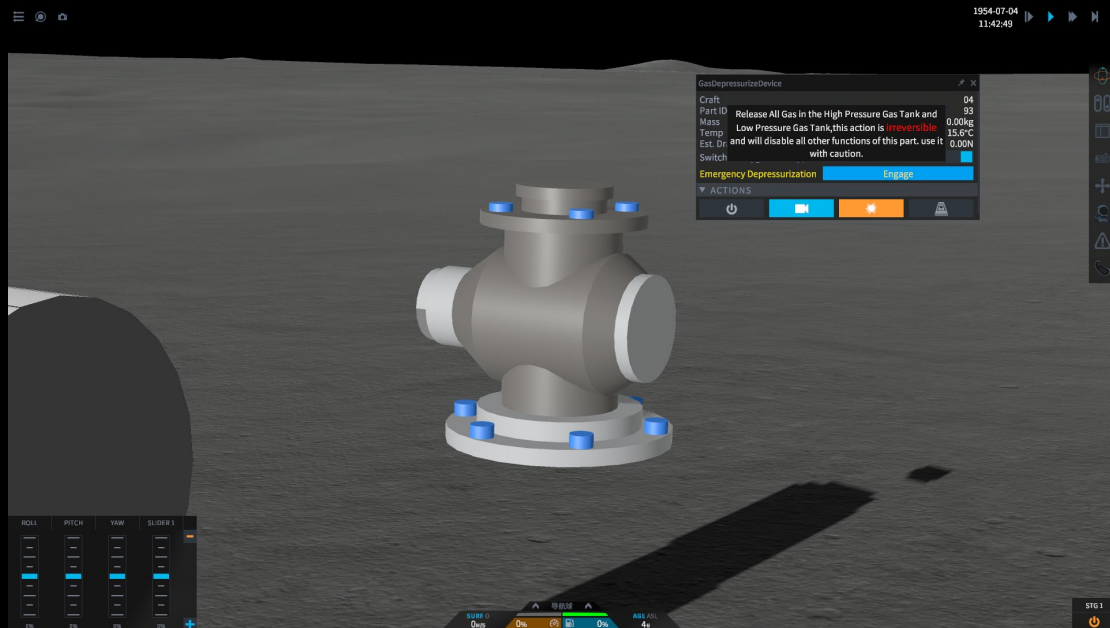
Here is a comparison of the size difference between Liquid oxygen and gas oxygen. You can see the Liquid oxygen takes much less room than the gas one. Yet, your crew will refuse to breathe liquid gas, so depressurization is required.



Gas Depressurize Device

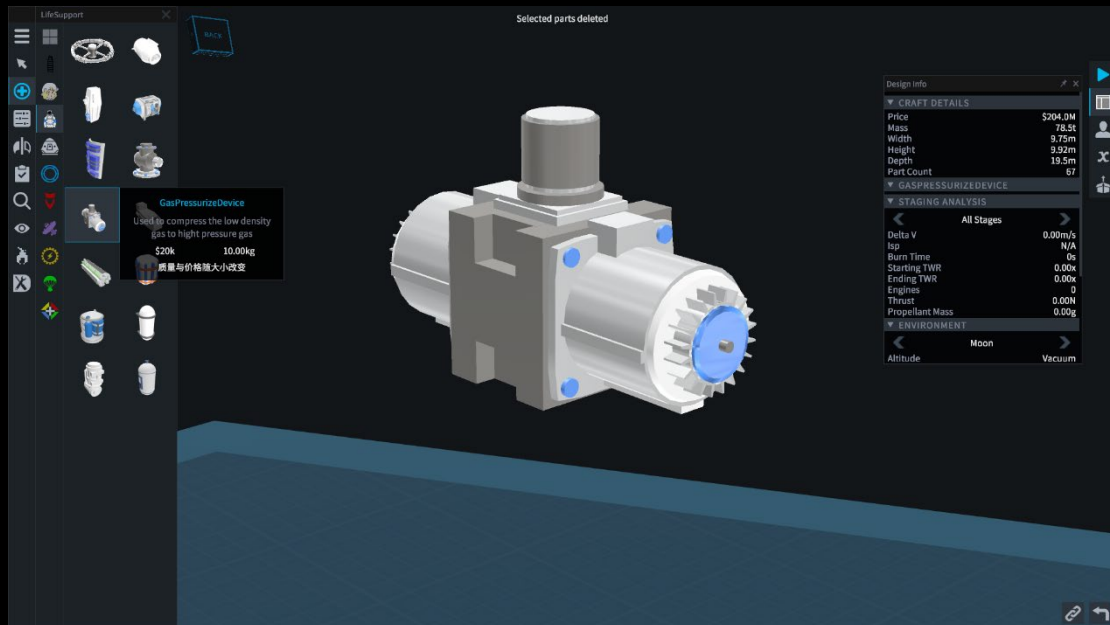
This part could turn HP gas or liquid gas(oxygen ,N2 and carben dioxide) into gas at 1 bar, there is an option to choose which gas type it should working with in the inspector panel.





This part also has a option called emergency depressurization, this action will release all gas in the craft, including high pressure gas and low pressure gas, after engaging , this part will be malfunctioning and can not function anymore.





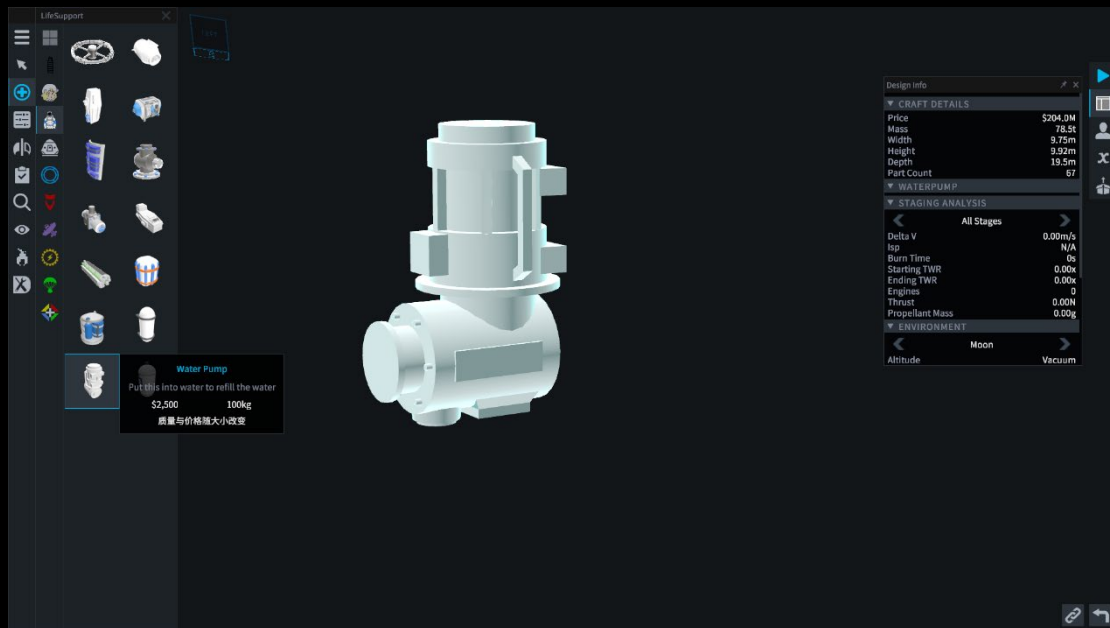
Gas Pressurize Device

Since there is already a part help “unpack” the HP gas, a part for compress low pressure gas to HP gas is no doubt necessary.

Unlike Gas Depressurize Device, this part requires electricity to function, turning oxygen, N2 or Carbon Dioxide into HP gas.

Like Gas Depressurize Device, you can also choose which gas type it should working with in the inspector panel.

Tips: One very common usage of this part is to packing the Carbon Dioxide if you don't want to simply throw them away.

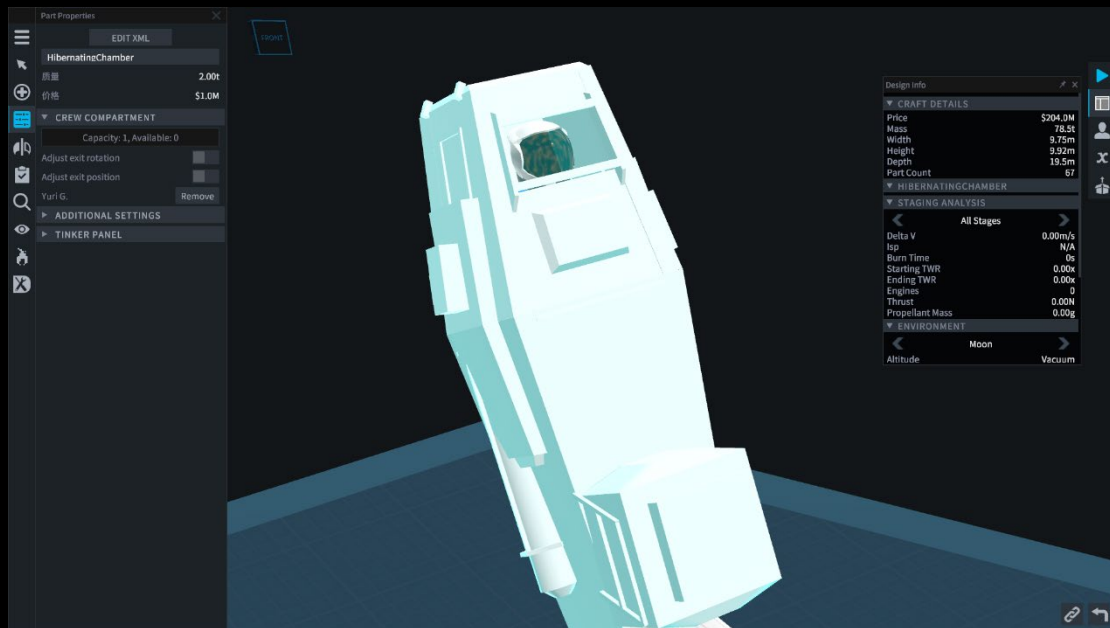


Water Pump

In stock game there is basically no way to refill the water, however, water plays an essential role in the whole cycle of life support, so when building your alien bases, bring this part and get it under water, then it will refill the water for the current craft.

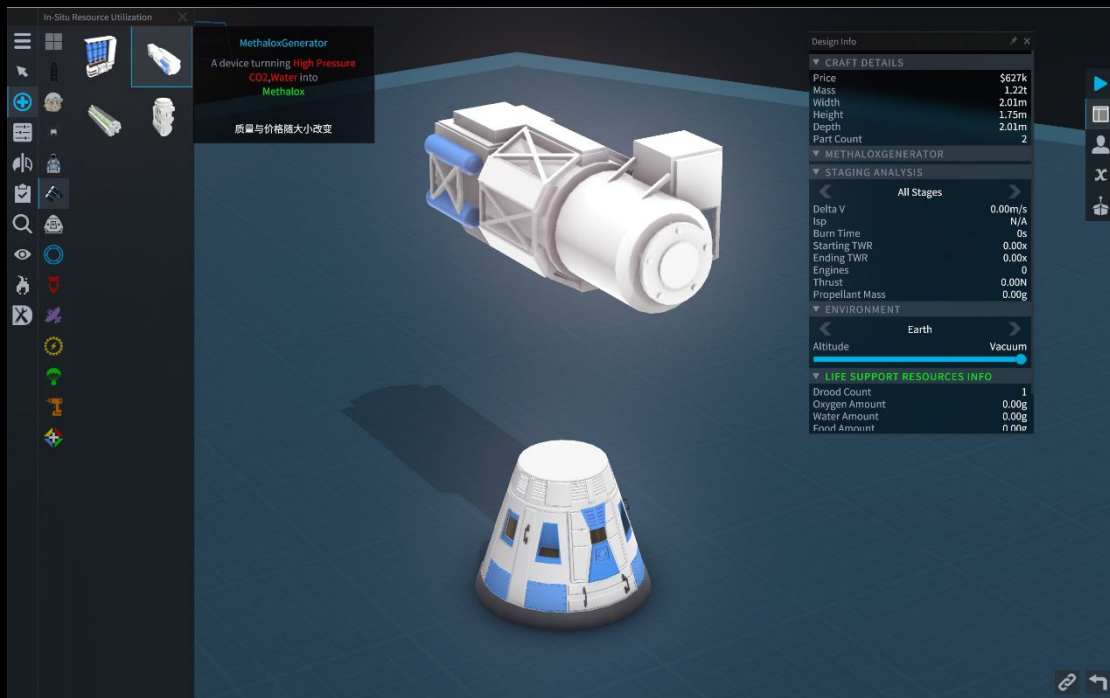
P.S

Fuel Cell which its fuel type is Hydrolox or Methalox could also function as a water pump, as long as it underwater and active.



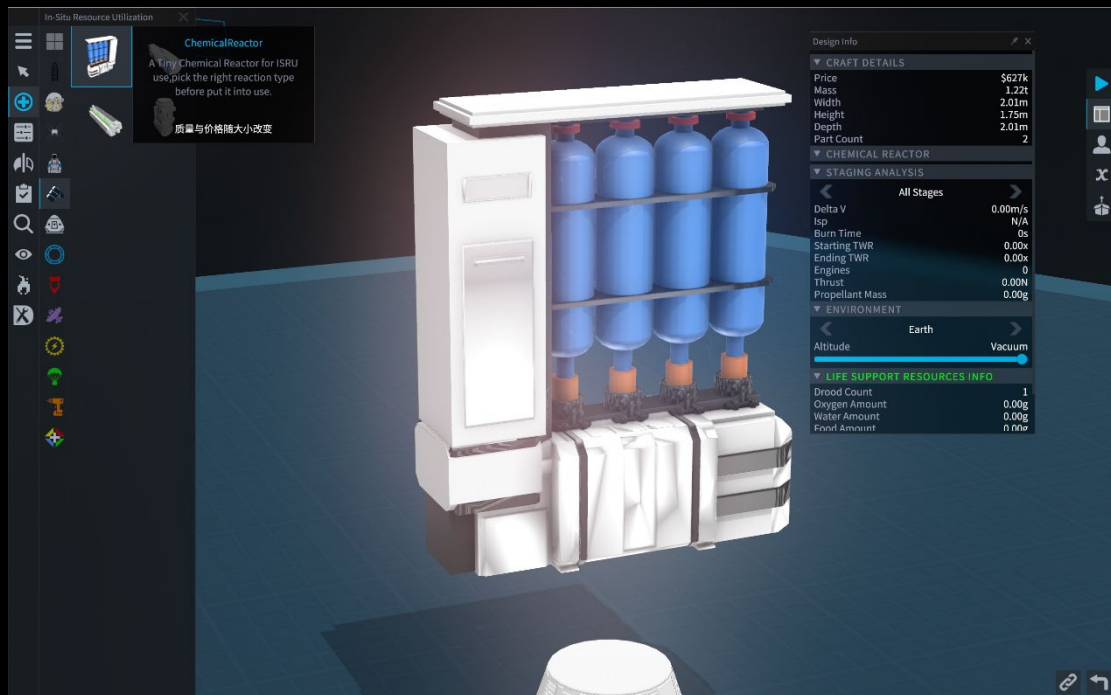
Hibernating Chamber

For those interstellar voyagers, or if you are simply too lazy to design a craft with a proper life support cycle system, introducing the Hibernating Chamber. When a Droid is inside a hibernating chamber with enough electricity, this Droid will not consume resources or generate waste. But be aware that this part consumes electricity like there is no tomorrow. Once out of power, the Droid inside will still consume resources and generate waste.



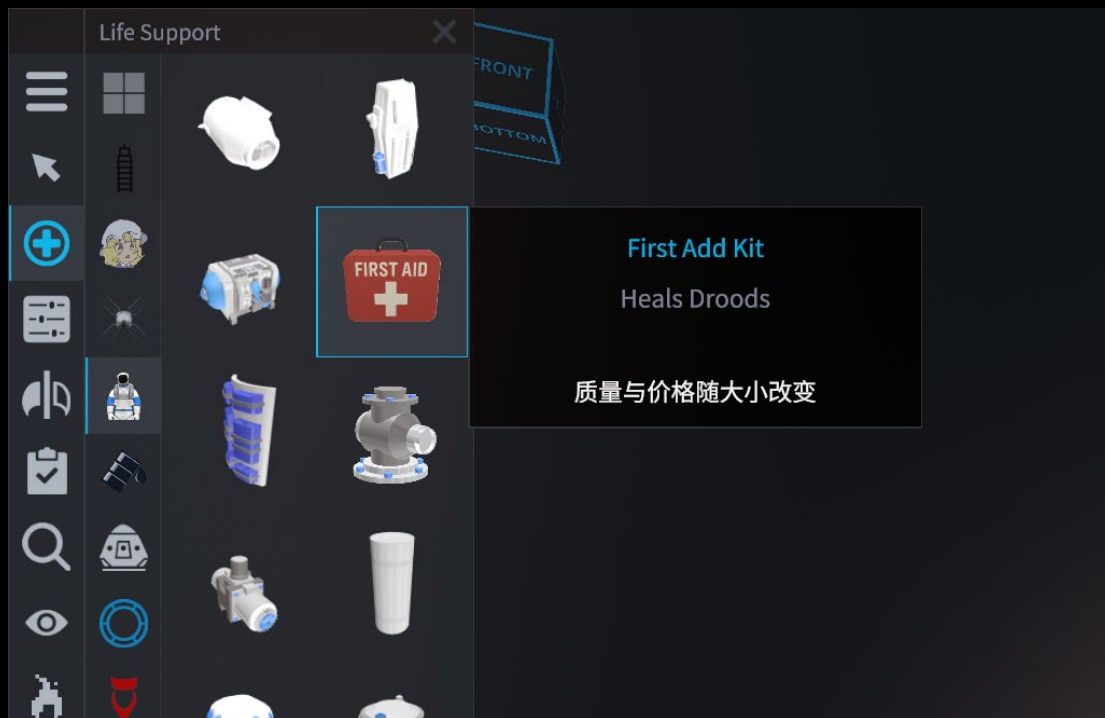
MethaloxGenerator

For those players who would like to build colonies in other planet , this part servers a specialized factory to produce Methalox only with High Pressure CO₂, and water.



Chemical Reactor

If you are willing to have more reaction type or build more fuel, this part lets you choose what to generate from Monopropellant, to Hydrolox and Methalox, however, please notice that this part's efficiency is not as high as MethaloxGenerator.

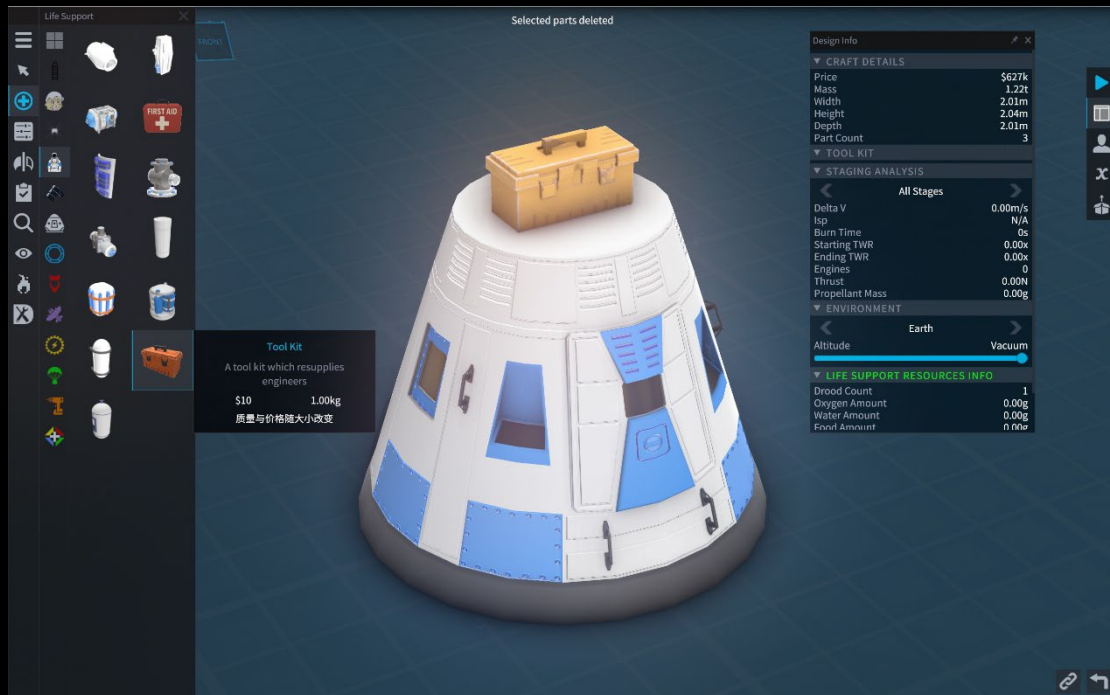


First Add Kit

Sometimes droids could get hurt during flight, if you still care about them, take a medic kit and heal your favorite droid.



By selecting a nearby drood and the first add kit, click “heal drood” will heal select drood.



Tool Kit

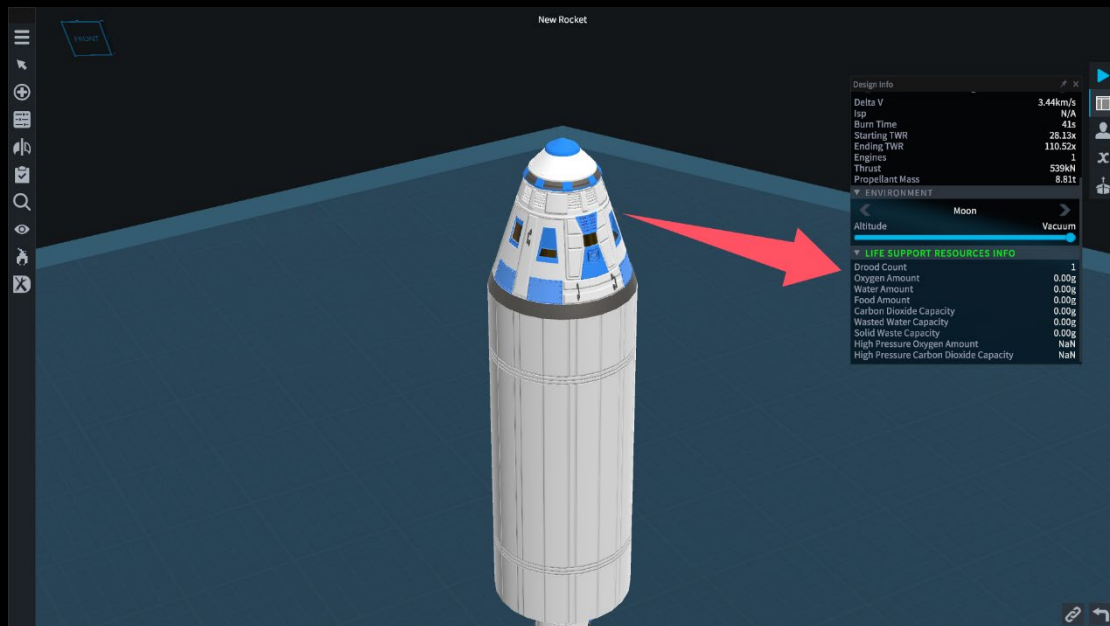
Engineer could fix parts using their repairing tools, when running out, you can use this part to refill them in flight



4.

4.

Designer



A new inspector panel showing life support resources data is added. You can check current craft's crew count, and resources amount.

One Important thing is that:

Oxygen, Water, Food, CO₂, Wasted Water, Solid Waste

Act like mono or battery, which means those corresponding fuel tanks do not need to connect to other part directly in order to make other parts function, as long as they are connected to the craft, it's good.

So how to design a crewed craft with life support enabled? Depending on one critical factor: TIME

If you are design a craft prepared for a rather short time trip, there is already a part for you called ResourcesPack, which integrated the nessesary resources for a single drood to live 24 hours.



If you would like to do a quick test and run to a stock craft or other's craft, attach this to the craft based on crew count or days, it should be fine for your crew to survive at least 24 hours

So, what if you want to try something more complicated? Like your own craft?

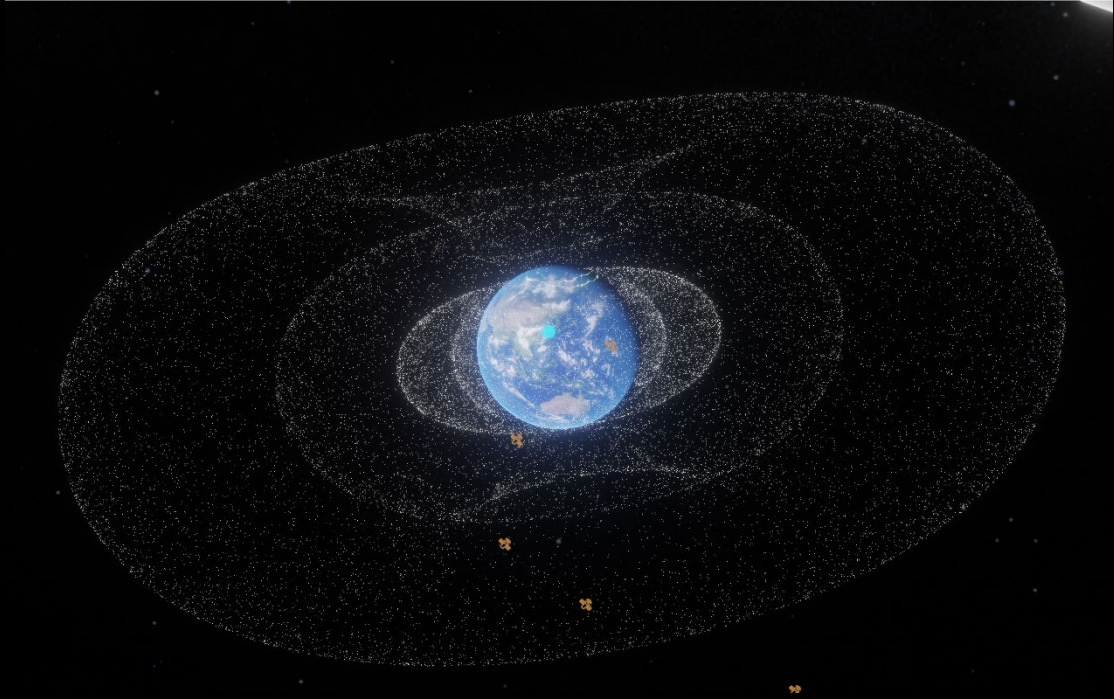
The best way to implement it is to imitate real spacecraft in human history. For instance, the Apollo Program's Saturn V utilized hydrolox fuel cells, and the ISS uses electrolysis of water to generate oxygen. You can do the same for your craft, again based on TIME.

Another factor you should take into account is the electricity system. All those life support related parts rely on enough electrical charge to function normally.

After all, try them out by yourself!

~~Definitely not because the mod creator is suck at building craft lol~~

5. Radiation Belt



Droodism Added Procedural Radiation Belt for some planets of juno system and Solar system.



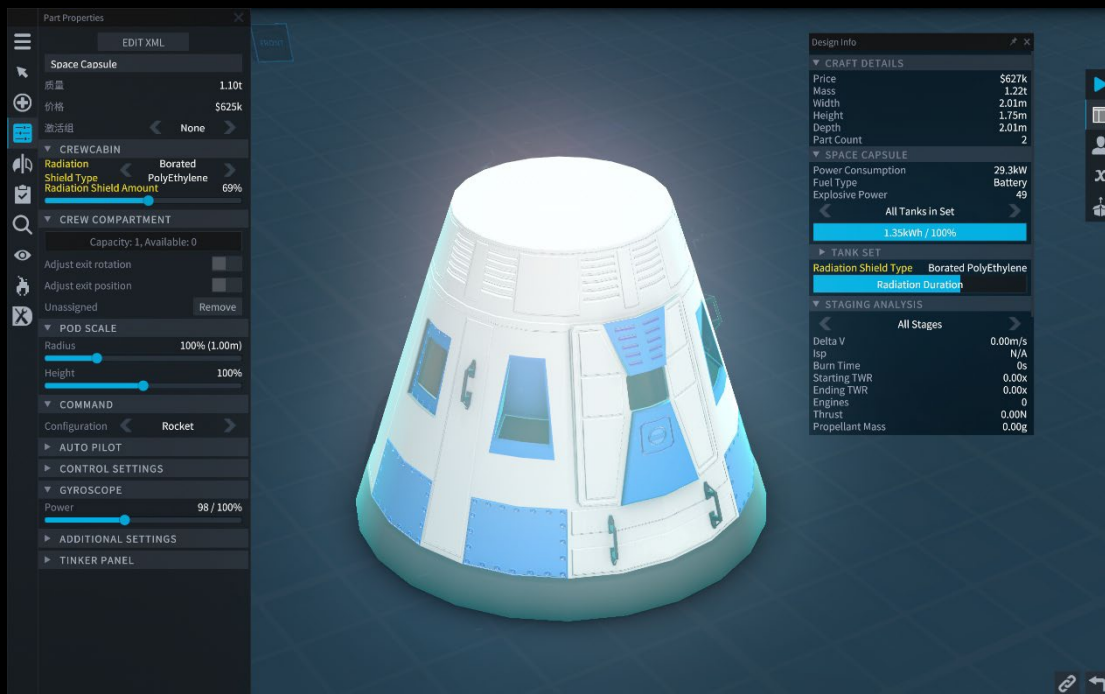
you can find radition belt inspector panel here
in Map Info.which shows the info of current
planet's radiation belt, also the toggle button
of visbility of radaiton belt



when your crews are inside the radiation belt, their radiation dose will begin to increasing, please be aware that **each crew's radiation dose is life long**. Once radiation dose rate or Cumulative radiation dose reached critical level, they will take damage and there is no way to stop them from taking damage now.

Radiation Protection

In order to protect your crew, you can config each crew compartment's radiation shield type and amount.



Each different radiation shield type has its own radiation cumulative reduction rate, exposing them in radiation belt will cause radiation shield duration loss regardless the crew count inside.

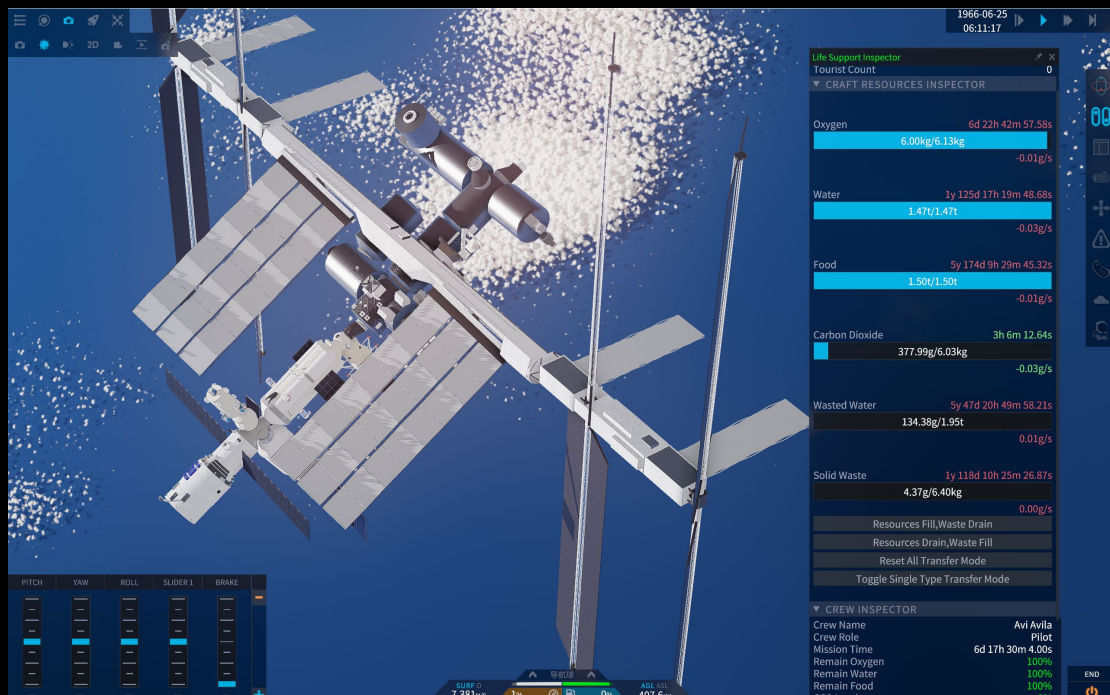
For Water and Liquid Hydrogen radiation shield, they can be refilled by using the corresponding fuel type in inspector panel



6. Flight and User Interface



A new flight inspector panel is added on the left side of the screen.



It shows quite a lot information, like crew count, but more importantly, the resources inspector is the main source.



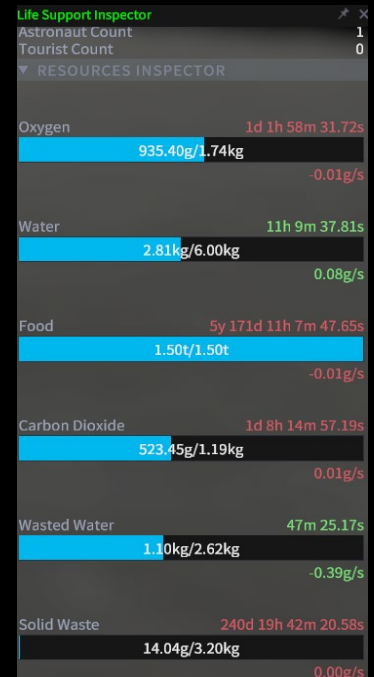
Taking the oxygen as an example, it will show the current amount of oxygen and its total capacity, current consuming speed, and how much time left.



However, when it turns green, that means it's being refilled, the time mean how long it takes to fill the oxygen up to full.

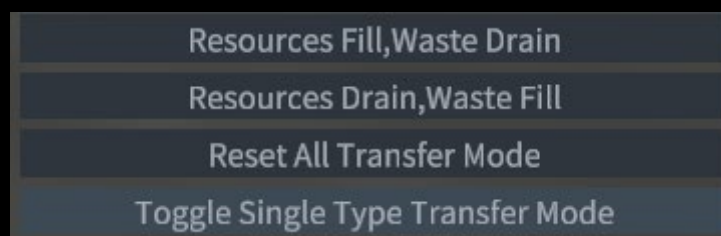
And for waste ,the color means

The opposite,you can see what happens When turning on sewage treating device .



Life Support Inspector		✕
Astronaut Count	1	
Tourist Count	0	
▼ RESOURCES INSPECTOR		
Oxygen	1d 1h 58m 31.72s	
935.40g/1.74kg		
	-0.01g/s	
Water	11h 9m 37.81s	
2.81kg/6.00kg		
	0.08g/s	
Food	5y 171d 11h 7m 47.65s	
1.50t/1.50t		
	-0.01g/s	
Carbon Dioxide	1d 8h 14m 57.19s	
523.45g/1.19kg		
	0.01g/s	
Wasted Water	47m 25.17s	
1.10kg/2.62kg		
	-0.39g/s	
Solid Waste	240d 19h 42m 20.58s	
14.04g/3.20kg		
	0.00g/s	

For some complex flights, transfer fuel, those resources and wastes are necessary, but hey! Don't you think it's a bit too annoying to select 6 or more fuel tanks and manually set up fuel transfer mode? So here is a cut corner for this:



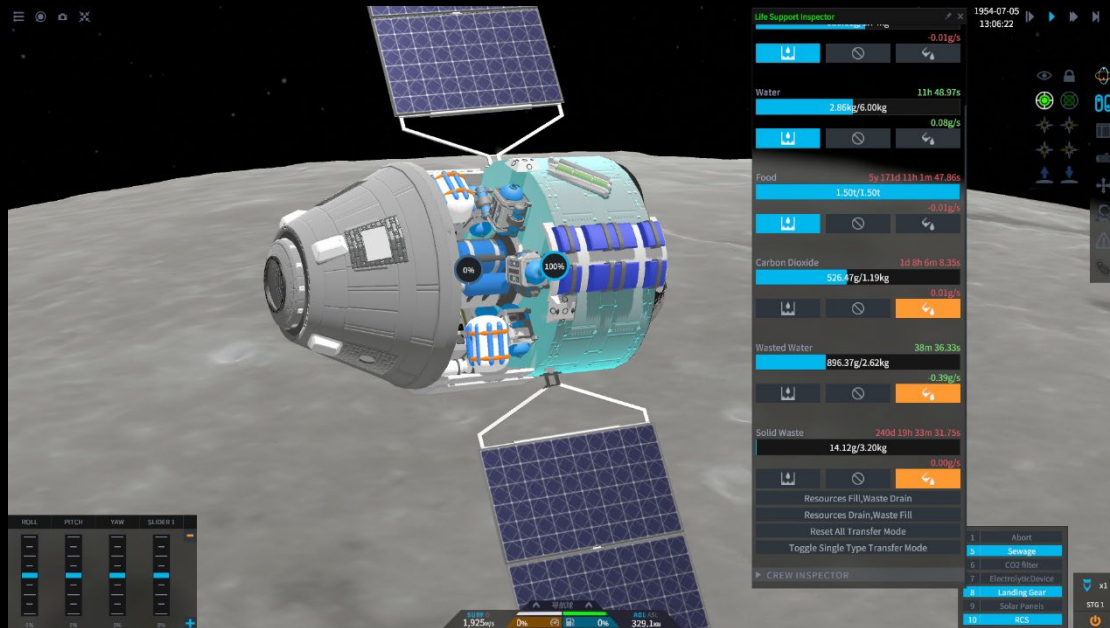
Resources Fill,Waste Drain
Resources Drain,Waste Fill
Reset All Transfer Mode
Toggle Single Type Transfer Mode

The first option will set oxygen ,food ,water fuel tanks to filling mode, while setting Carbon Dioxide,

wasted water and solid waste fuel tanks to
draining mode

The second option does the counter operation to
the first one

The third option will reset all fuel transfer mode
back to none



If you need more precise operations, the fourth option allow you to set each fuel type's fuel transfer mode individually.

▼ CREW INSPECTOR		
Crew Name		Yuri G.
Crew Role		Pilot
Mission Time		32.00s
Remain Oxygen	Using External Oxygen	
Remain Water	100%	
Remain Food	100%	
CO2 Level	Using External Oxygen	
Wasted Water Level	0%	
Solid Waste Level	0%	
Radiation Dose		17.1354 rad
Radiation Stats	nominal	
Radiation Rate		0.00 rad/h
Radiation Rate Stats	nominal	

And the last part of the inspector:showing crew's name,Life Support Resouces,Radiation Dose , and corresponding mission time.

Inspector still works in EVA, but you can not set fuel transfer mode, obviously.

6. Credits

This mod can not be done without the help of following players:

Models by w2semail, TN920, DeepSpaceIndustries

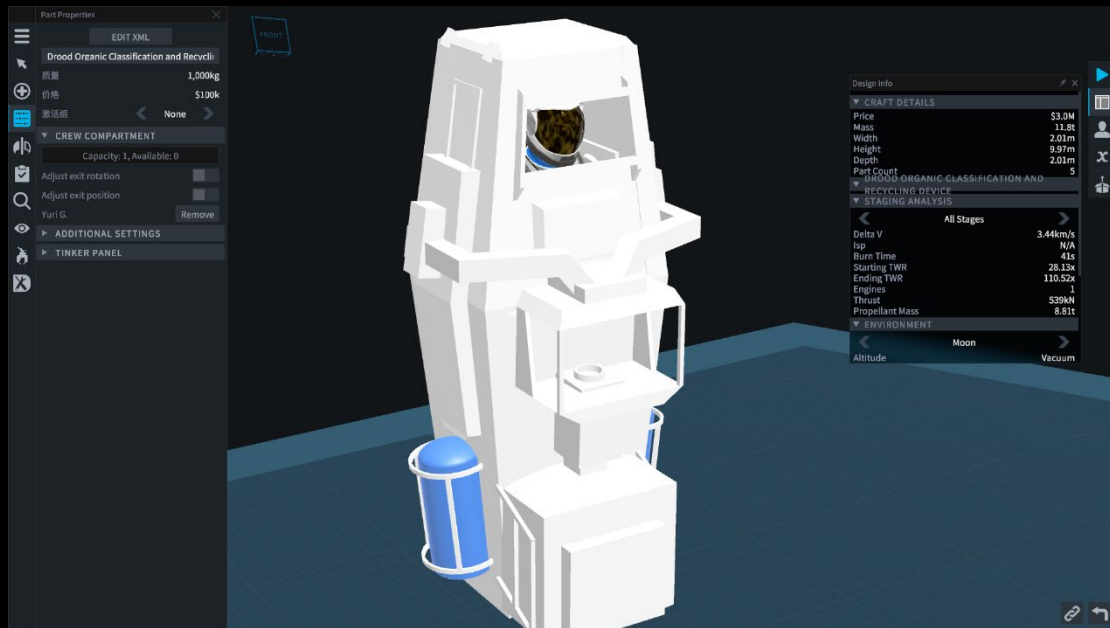
Mod Thumbnail by TN920

Tesing Players: Charodey, GreenGalaxyAeroSpace

And YOU!

Thanks for reading this manual,
and have fun with Droidism!

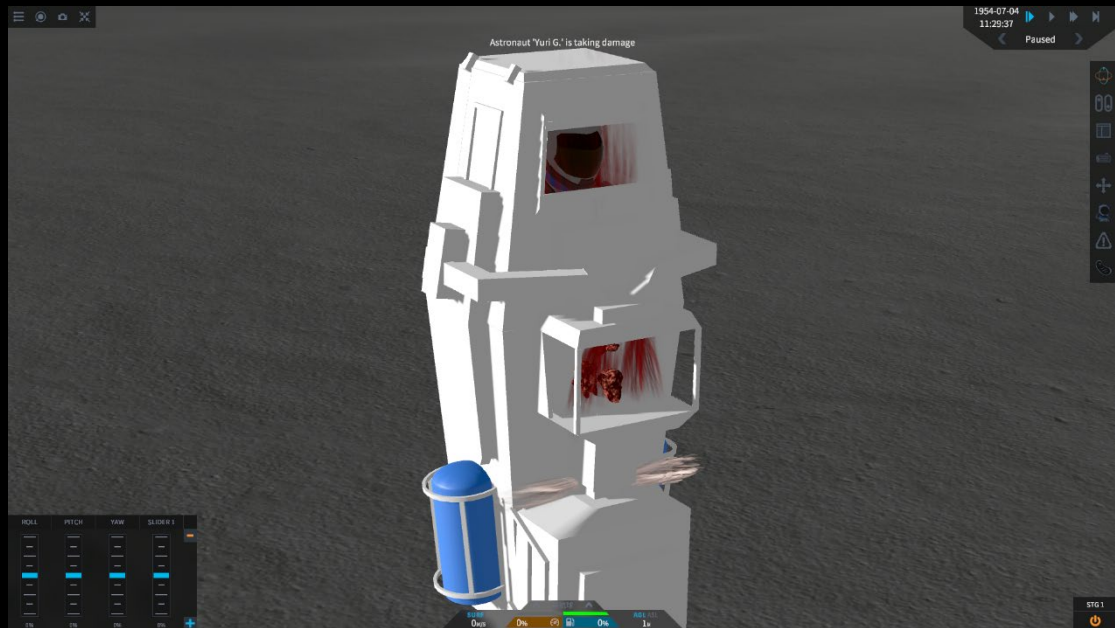
END OF THE MANUAL



Drood Organic Classification and Recycling Device

What would you do if running out of resources in the voyage? Or What could you do when see one of you least favorite crew sharing the precious food with others?

Time to tell them, for the greater good, there needs to be sacrifice.



This part will drains all resources in current drood and kills this crew, then its high efficiency classification system will extract every drop of water from this crew's blood and flesh.

Since you can not heal a drood's health in flight, so, instead of letting an injured drood waste your resources, why not utilize this part just kill this crew and recycle the resources from him?